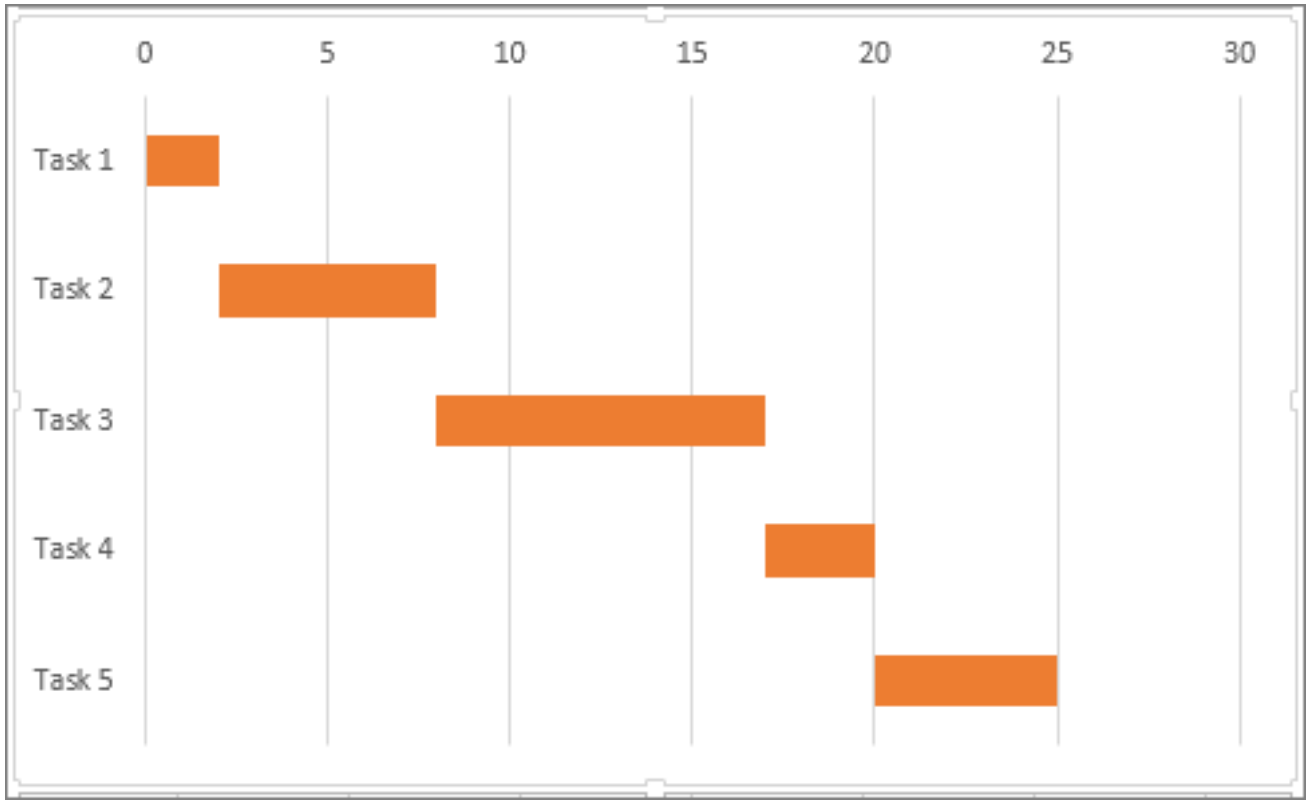
CSE 407

Gantt Chart

Gantt Chart Overview



A Gantt Chart is basically a horizontal bar chart which is used as a Production Control Tool

Gantt chart provides a graphical illustration of a **schedule** that can be used to plan, coordinate and track tasks in a project

It's a visual representation that gives an instant overview of the status of a project

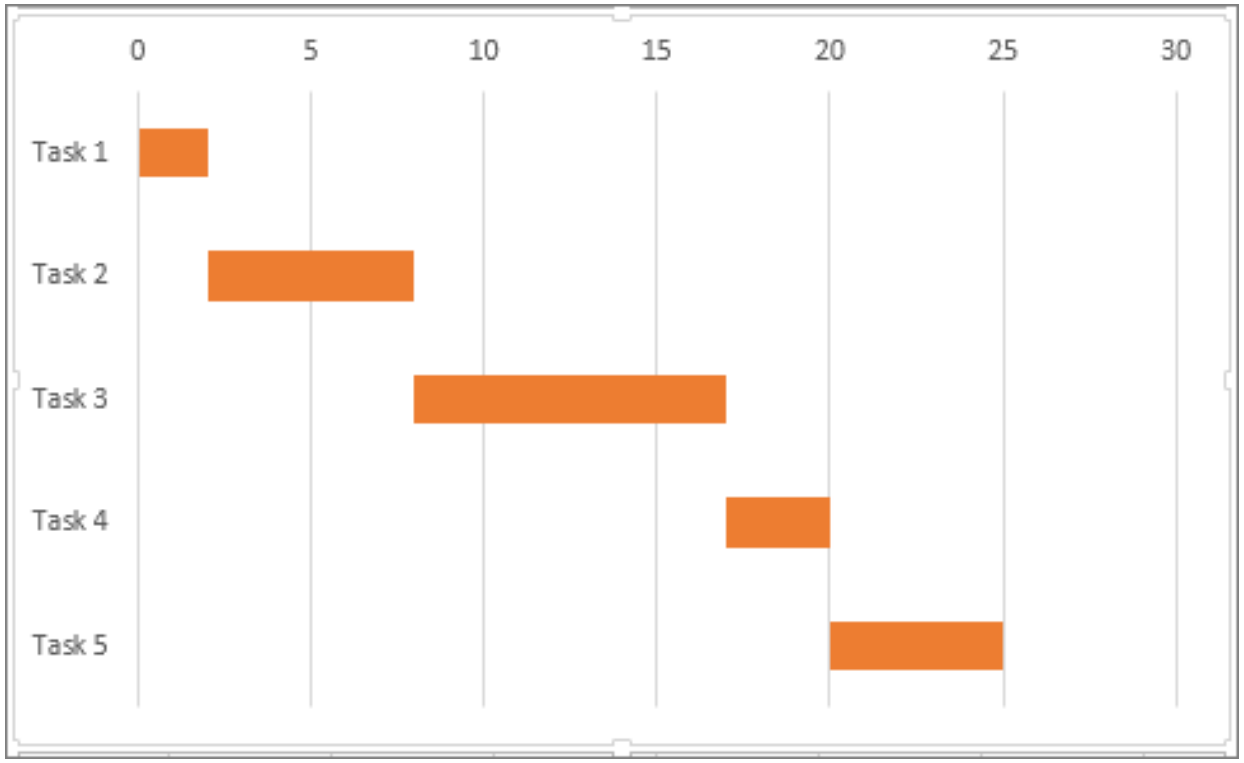
What are Gantt Chart used for?

- **Progress monitoring**: Project managers can see if individual tasks are completed on time and adjust the project schedule. The charts also show which goals were met on schedule, helping managers gauge employee productivity.
- **Project planning**: Managers can set deadlines, milestones and schedules for various project components.
- **Resource management**: Project planners can coordinate resource allocation with the project schedule. Managers can see the amount of time each process takes and designate resources accordingly.

What are Gantt Chart used for?

- **Waterfall** and **Agile** both methodologies use Gantt Charts
- Mainly used in **Waterfall** because a linear plan is followed
- Gantt charts can be useful in **Agile** to compare an old plan to a proposed change and to see what effect the change has on the overall plan

How to make Gantt Chart



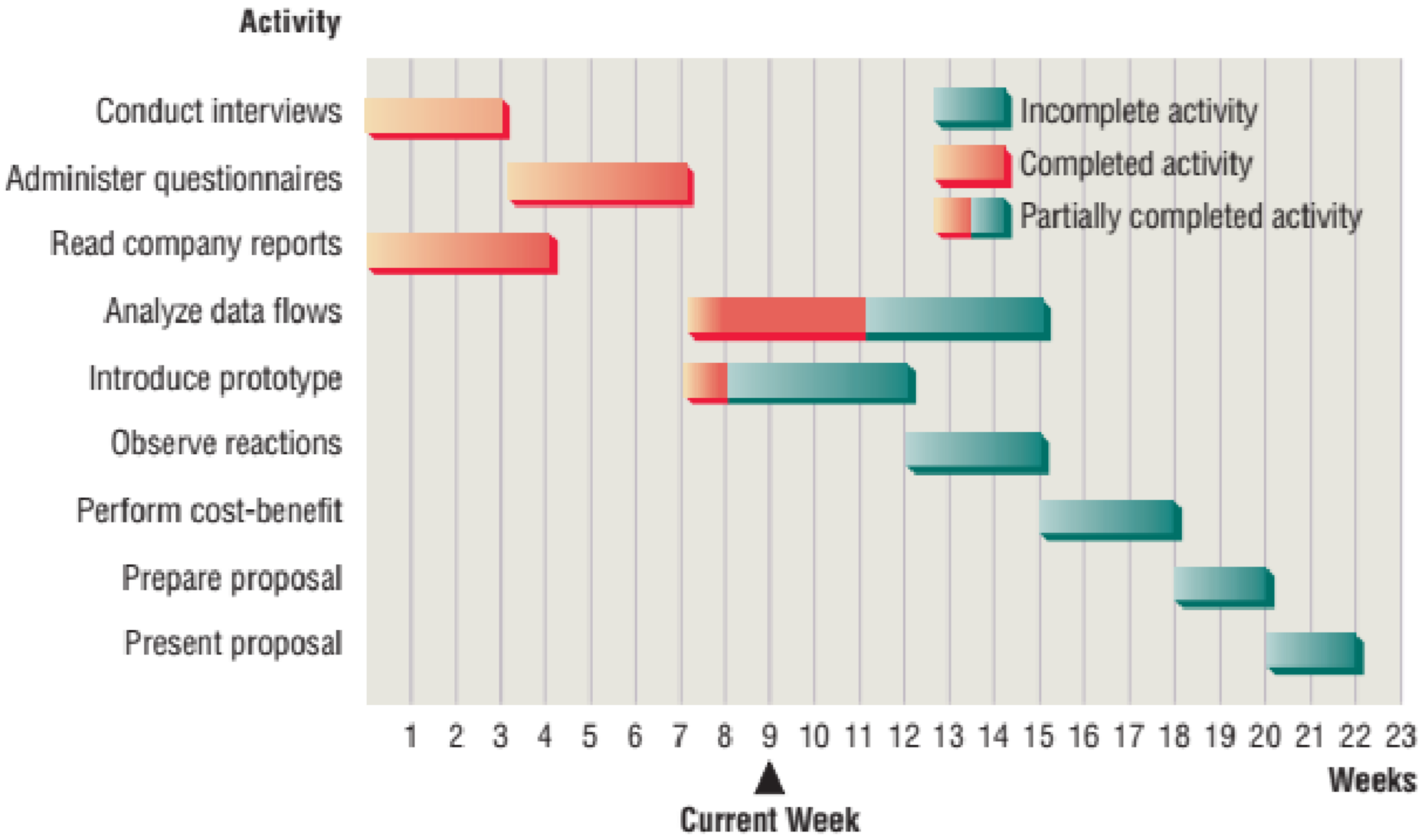
- **Horizontal Axis** represents total project time span broken into increments
- **Vertical Axis** represents project tasks
- **Horizontal bars** of varying lengths represent the sequences, timing, and time span for each task
- **Colored portion** of bars indicate amount of work completed

Example

Activity	Detailed Activity	Weeks Required
Data gathering	Conduct interviews	3
	Administer questionnaires	4
	Read company reports	4
	Introduce prototype	5
	Observe reactions to prototype	3
Data flow and decision analysis	Analyze data flow	8
Proposal preparation	Perform cost-benefit analysis	3
	Prepare proposal	2
	Present proposal	2



Example


Activity
Conduct interviews
Administer questionnaires
Read company reports
Analyze data flows
Introduce prototype
Observe reactions
Perform cost-benefit
Prepare proposal
Present proposal
Incomplete activity
Completed activity
Partially completed activity
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Weeks
Current Week

Benefits of Gantt Charts

- **Efficiency**: Gantt charts help project managers calculate realistic project completion times and set goals based on available resources. Both of these advantages increase productivity.
- **Teamwork**: Team members have access to the same information, keeping everyone informed of a project's progress. Because of this, it's possible to hold all team members, including remote workers, accountable for their tasks. Team members can reference the chart and establish roles and responsibilities.
- **Tracking**: Gantt charts allow project planners and team members to analyze workflows for constraints and adjust their work accordingly. Gantt charts help project managers track benchmarks and tasks throughout the project process. Team members can easily visualize which elements may be missing from the Gantt chart.

Benefits of Gantt Charts

- **Versatility**: Gantt charts help teams balance multiple projects at once. They prompt managers to estimate which resources are needed when. They are also easy to change.
- **Visualization**: Gantt charts provide a project roadmap at a glance, allowing for easier management, monitoring and organization of project components. The chart gives a holistic view of the project timeline and tasks. It also provides high-level visibility into the who, what, when and where of a project. Employees can quickly see the status of the project or a project phase.

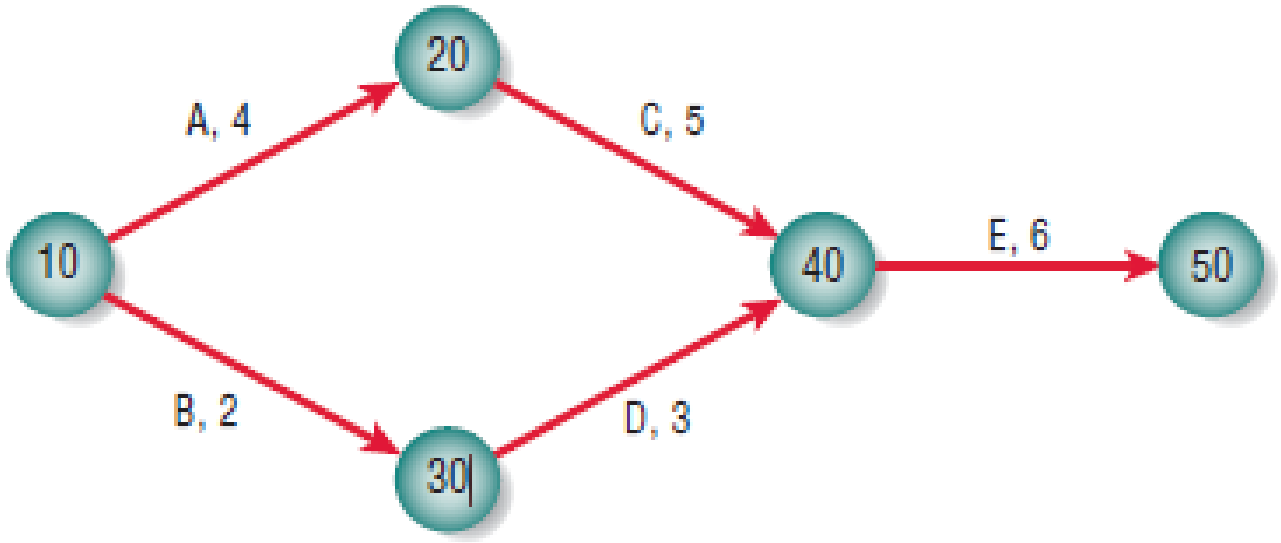
Limitations of Gantt Charts

- **Lack of dependencies**: Gantt charts don't indicate task dependencies or critical paths of projects in detail. Users cannot tell how one task falling behind schedule affects other tasks. The PERT chart, another popular project management method, is designed to do this.
- **Lack of specificity**: Gantt charts are good for providing a high-level view of project workflows. However, they can leave some things to interpretation. This is especially true for large-scale IT projects, such as DevOps Certain details about priorities, deadlines and expectations may be left out of the high-level view of complex projects.
- **Quantity of work**: A third issue with Gantt charts is that the task bar doesn't show the quantity of work each task requires. It may show a specific length of time, which roughly implies a certain amount of resources. However, it does not designate the resources. That creates ambiguity and can lead to resource contention.

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PERT Diagram

PERT Diagram Overview

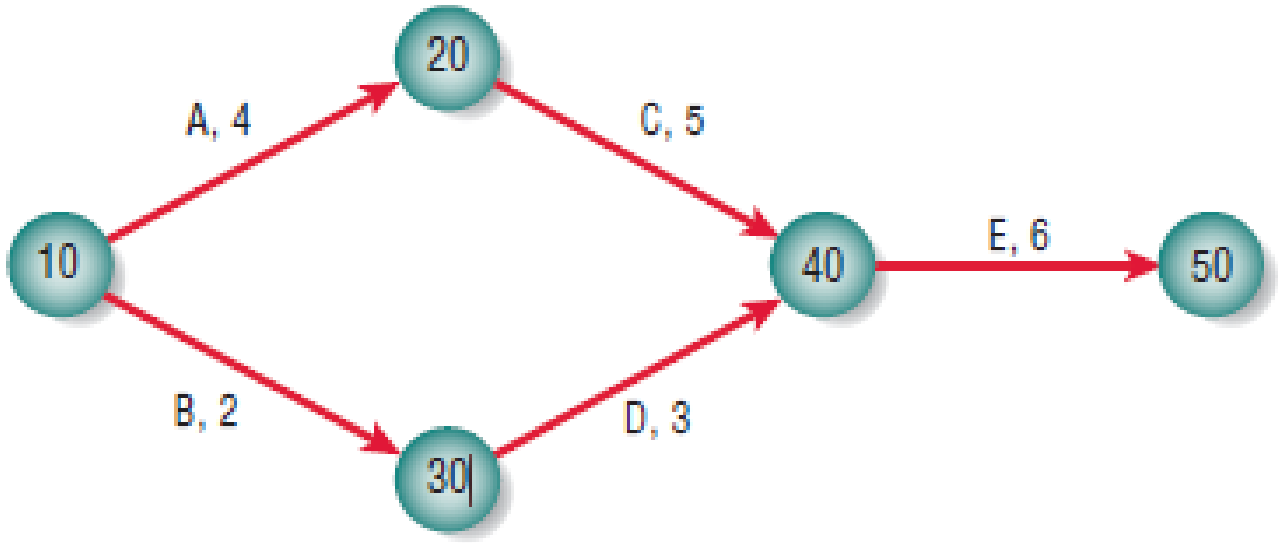


PERT is an acronym for **Program Evaluation and Review Technique**

A project is represented by a network of **nodes** and **arrows** that are evaluated to determine the **critical activities**, improve the schedule if necessary, and review progress once the project is undertaken

It is useful when activities can be done in **parallel** rather than in sequence

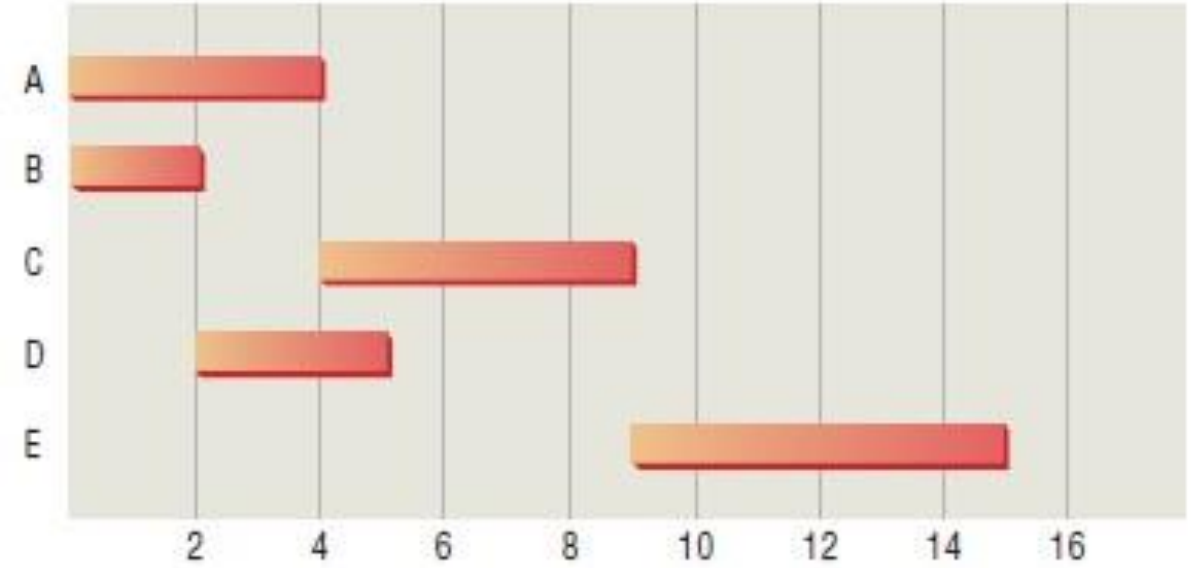
PERT Diagram Overview

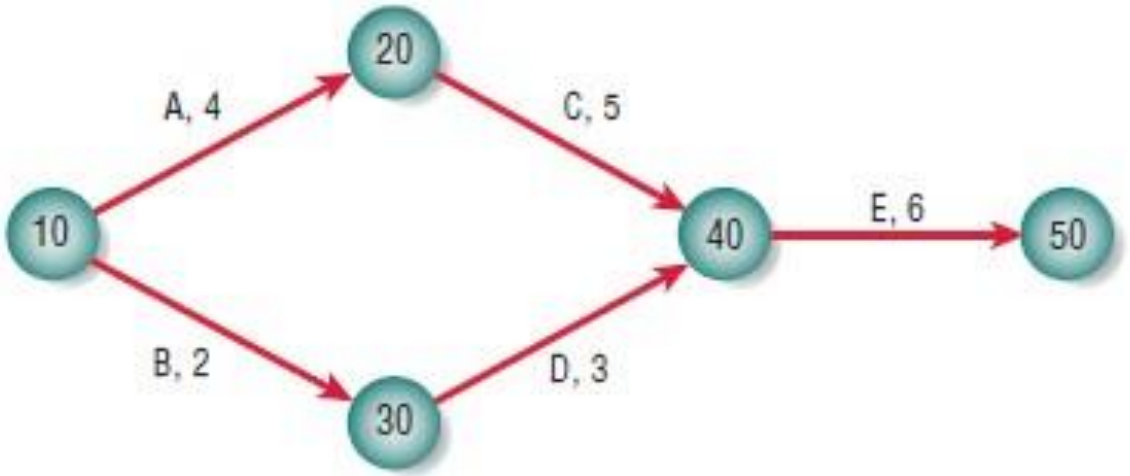


Length of arrow has no relationship with activity duration. They are arbritary

Circles or nodes are called **events**, and can be identified by numbers, letters etc.

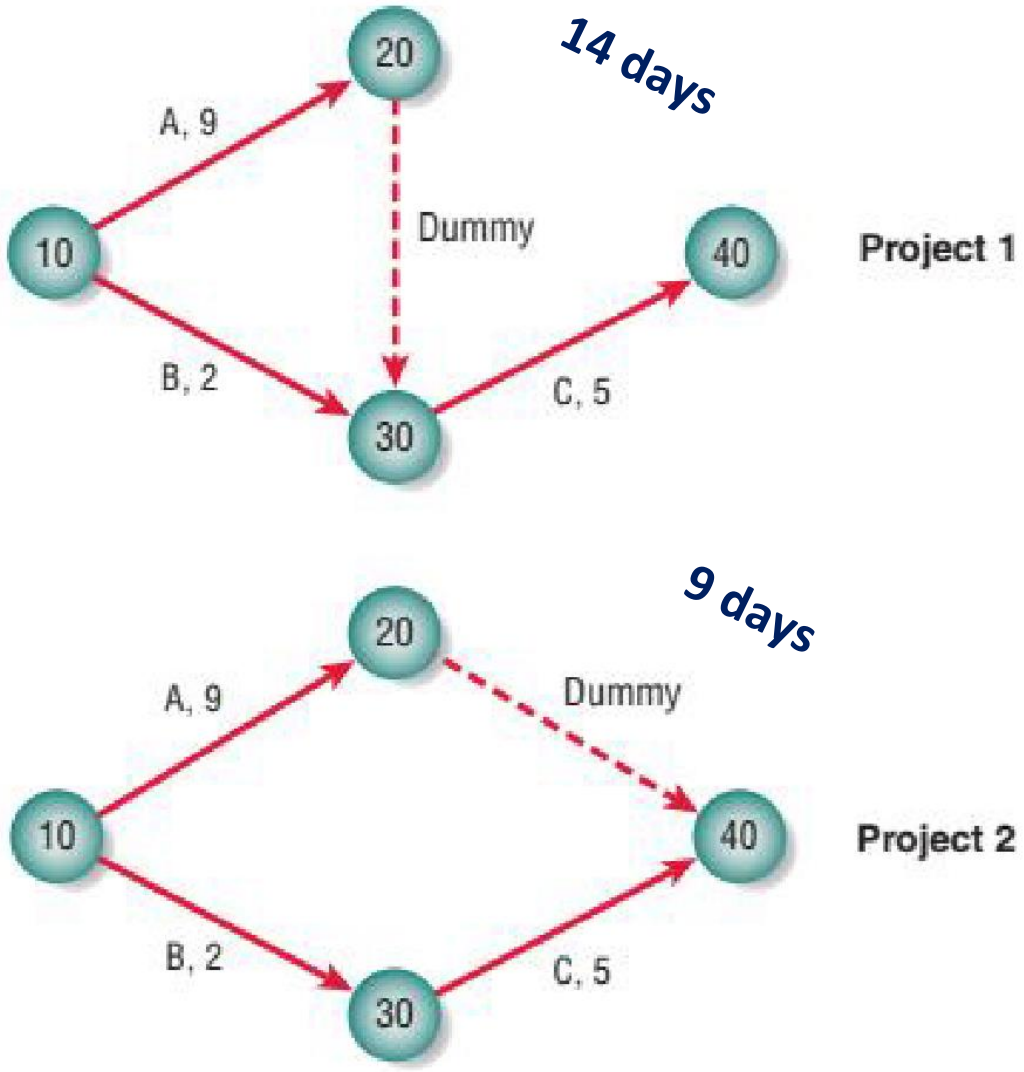
Gantt Chart to PERT Diagram





The activities expressed as bars in the Gantt chart are represented by arrows in the PERT diagram

Dummy activities for precedence

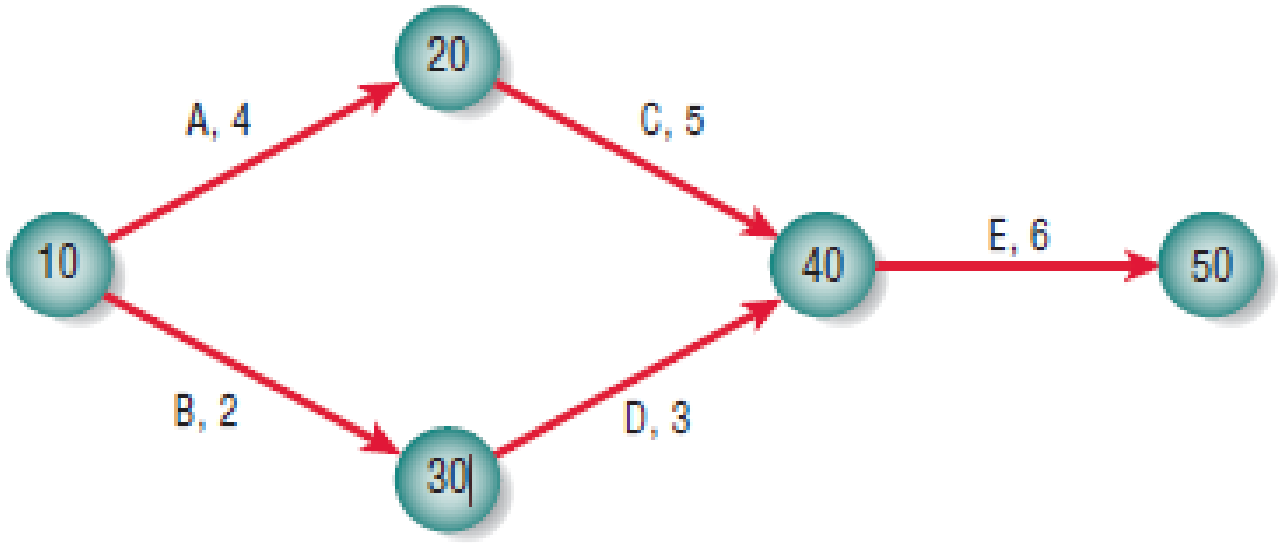


Dummy activities are introduced to preserve project logic and precedence

In project 1 activity C can only be started if both A and B are finished, because all arrows coming into a node must be completed before leaving the node

In project 2 activity C requires only activity B's completion and can therefore be under way while activity A is still taking place. Node 20 can be deleted

Critical Path



10-20-40-50 is the **Critical Path**

Critical Path is defined as the path that will cause the whole project to fall behind if even one day's delay is encountered on it.

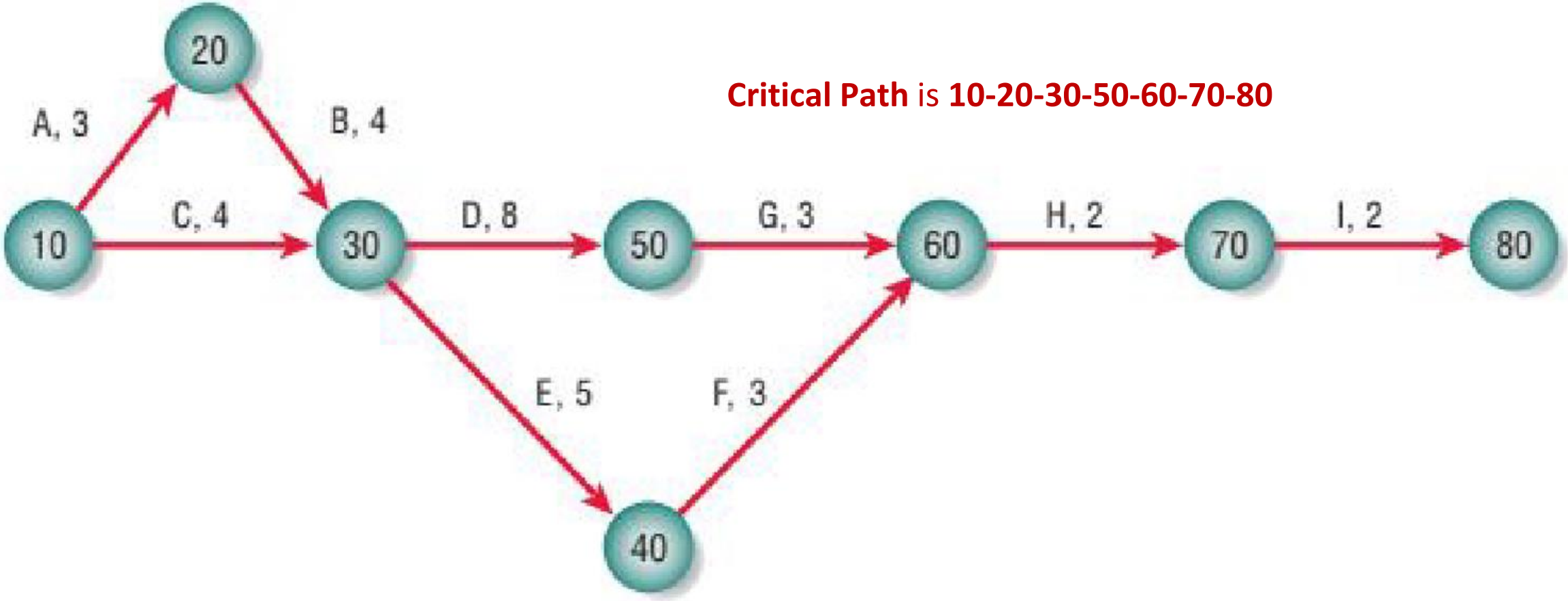
It is the longest path of the project

The leeway to fall behind somewhat on noncritical paths is called **slack time**.

Critical Path Example

Activity		Predecessor	Duration
A	Conduct interviews	None	3
B	Administer questionnaires	A	4
C	Read company reports	None	4
D	Analyze data flow	B, C	8
E	Introduce prototype	B, C	5
F	Observe reactions to prototype	E	3
G	Perform cost-benefit analysis	D	3
H	Prepare proposal	F, G	2
I	Present proposal	H	2

Critical Path Example



Critical Path is **10-20-30-50-60-70-80**

Using PERT Diagram over Gantt Chart

1. Easy identification of the order of precedence.
2. Easy identification of the critical path and thus critical activities.
3. Easy determination of slack time.